

ACIDS, BASES AND SALTS

Teacher Lesson Guide

1. Metadata

Topic	Acids, Bases and Salts
Level	Class X
Lesson Length	60 minutes

Scope Note

Focus on classification of salts, pH behaviour based on parent acid/base strength, and interpretation of experimental observations.

2. Learning Objectives

- LO1: Identify the acids and bases used to form given salts and classify salts into families.
- LO2: Explain how the strength of the parent acid and base determines the pH of the salt solution.
- LO3: Interpret experimental pH data to categorize salts as acidic, basic or neutral.
- LO4: Describe the pH scale and its significance for classifying solutions.
- LO5: Recognise sodium chloride as a neutral salt and relate it to everyday uses.

3. Prerequisite Knowledge

- Understanding that acids release H^+ ions and bases release OH^- ions in aqueous solution.
- Difference between strong and weak acids/bases.
- Use of litmus paper and pH paper to test acidity or basicity.
- Basic idea that salts are formed by neutralisation of an acid with a base.

4. Key Terms

Term	Definition
Salt	A compound formed when an acid reacts with a base, consisting of the cation from the base and the anion from the acid.
Acid	Substance that releases H^+ ions in water.
Base	Substance that releases OH^- ions in water.
Strong acid	An acid that dissociates completely in water, e.g., HCl.
Weak base	A base that does not dissociate completely in water, e.g., ammonia.
pH	A numerical scale (0-14) that indicates the concentration of hydrogen ions; lower values are acidic, higher values are basic.
Neutral salt	A salt formed from a strong acid and a strong base; its aqueous solution has $pH \approx 7$.

Term	Definition
Acidic salt	A salt whose solution has $\text{pH} < 7$, typically formed from a strong acid and a weak base.
Basic salt	A salt whose solution has $\text{pH} > 7$, typically formed from a weak acid and a strong base.
Salt family	Group of salts sharing the same positive ion (cation) or the same negative ion (anion).

5. Key Points

- Salts sharing the same cation or anion belong to the same family.
- The pH scale (0-14) indicates whether a solution is acidic, neutral, or basic.
- A salt's pH depends on whether its parent acid and base are strong or weak.
- Students test solubility, observe litmus colour change, and record pH of various salts.
- Sodium chloride, formed from a strong acid and strong base, is neutral and widely used.

6. Real-World Applications

Sodium Chloride in Food Preservation

Common salt inhibits microbial growth by reducing water activity, keeping food safe and tasty.

Think about: Why does adding NaCl to meat help it stay fresh longer?

Expected answer: NaCl draws out water from microbial cells, creating an environment with low water activity that prevents bacterial growth.

Suggested time: 3 minutes

pH of Household Cleaning Solutions

Many cleaning agents are either acidic (e.g., vinegar) or basic (e.g., bleach) to enhance their effectiveness.

Think about: A bathroom cleaner turns blue litmus red. What does this indicate about its pH and why is that useful for cleaning?

Expected answer: The cleaner is acidic ($\text{pH} < 7$); acids help dissolve mineral deposits and soap scum by reacting with basic residues.

Suggested time: 4 minutes

7. Teacher Schedule

Time	Stage	Teacher Direction	Student Action
0-5 min	Introduction	Introduce the lesson topic and learning objectives. Use brief retrieval questioning to activate relevant prerequisite knowledge.	Respond to retrieval questions and review the learning objectives.
5-13 min	Teaching Phase 1	Guide students to match each salt with its parent acid and base and group salts sharing the same ion.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
13-23 min	Teaching Phase 2	Emphasise strong-acid/strong-base $\rightarrow$ neutral, strong-acid/weak-base $\rightarrow$ acidic, weak-acid/strong-base $\rightarrow$ basic.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.

Time	Stage	Teacher Direction	Student Action
23-35 min	Teaching Phase 3	Analyse Table 2.4 results and link pH values to the nature of the parent acid/base.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
35-42 min	Teaching Phase 4	Highlight the 0-14 scale, meaning of pH 7, and how ion concentration relates to pH. Use the real-world application 'pH of Household Cleaning Solutions': Many cleaning agents are either acidic (e.g., vinegar) or basic (e.g., bleach) to enhance their effectiveness. Ask students: A bathroom cleaner turns blue litmus red. What does this indicate about its pH and why is that useful for cleaning?	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
42-47 min	Teaching Phase 5	Connect the concept of neutral salts to common table salt and its historical relevance. Use the real-world application 'Sodium Chloride in Food Preservation': Common salt inhibits microbial growth by reducing water activity, keeping food safe and tasty. Ask students: Why does adding NaCl to meat help it stay fresh longer?	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
47-50 min	Consolidation / Review	Use the remaining time to revisit key points, check understanding, clarify difficult concepts or practise a short retrieval question.	Review key learning and ask questions about any remaining areas of uncertainty.
50-55 min	Question Phase	Students complete Q1, Q2, Q3, Q4, Q5. Review answers and address remaining gaps in understanding.	Answer the lesson questions and review responses.
55-60 min	Reflection / Review	Review the learning objectives and identify areas requiring further review.	Review the learning objectives and identify any topic requiring further review.

8. Activities

Suggested Activities If Time Available

Identify Parent Acids and Bases for Given Salts (12 min)

Using the list of salts from Activity 2.14, students write the chemical formula of the acid and base that combine to form each salt, then group the salts into families based on common cations or anions.

Predict Salt Solution pH from Parent Strengths (14 min)

Students predict whether each salt from Activity 2.14 will be acidic, basic, or neutral by analysing the strength (strong/weak) of its parent acid and base, then compare predictions with the measured pH values.

9. Questions and Answers

Q1. What is a salt family and give an example of two salts belonging to the same family.

Answer: A salt family consists of salts that share the same cation or the same anion. Example: NaCl and Na₂SO₄ are both sodium salts; NaCl and KCl are both chloride salts.

Q2. Predict whether a salt formed from a strong acid and a weak base will be acidic, neutral or basic, and explain why.

Answer: It will be acidic ($\text{pH} < 7$) because the conjugate acid of the weak base hydrolyses in water, releasing H^+ ions.

Q3. In Activity 2.14, sodium acetate gave a pH of about 9. What does this indicate about the nature of the salt and its parent acid/base?

Answer: Sodium acetate is a basic salt; it is formed from a strong base (NaOH) and a weak acid (acetic acid).

Q4. State the pH value that defines a neutral solution and the ranges for acidic and basic solutions.

Answer: Neutral solution: $\text{pH} = 7$; acidic solution: $\text{pH} < 7$; basic solution: $\text{pH} > 7$.

Q5. Why is sodium chloride considered a neutral salt?

Answer: Because it is formed from a strong acid (HCl) and a strong base (NaOH), its aqueous solution has a pH close to 7.